

Table 1: Effect of successive diagnose-refine rounds with IntentTrace. mean specification-alignment precision (P), recall (R), and F_1 before refinement and after one round per diagnostic approach. Bold marks the highest P, R, and F_1 within each dataset-LLM setting. The values averaged over more than one run are provided with their standard deviation values.

Refinement	Industry						Paged					
	Gemini			Qwen			Gemini			Qwen		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
No Refinement	0.32 $\pm$ 0.004	0.44 $\pm$ 0.001	0.36 $\pm$ 0.004	0.29 $\pm$ 0.003	0.46 $\pm$ 0.003	0.32 $\pm$ 0.001	0.41 $\pm$ 0.003	0.47 $\pm$ 0.003	0.43 $\pm$ 0.002	0.38 $\pm$ 0.002	0.47 $\pm$ 0.003	0.41 $\pm$ 0.002
1st Round	0.71	0.82	0.72	0.55 $\pm$ 0.010	0.81 $\pm$ 0.012	0.61 $\pm$ 0.009	0.86	0.85	0.83	0.72 $\pm$ 0.011	0.81 $\pm$ 0.006	0.74 $\pm$ 0.006
2nd Round	0.83	0.88	0.82	0.65 $\pm$ 0.026	0.86 $\pm$ 0.028	0.69 $\pm$ 0.026	0.94	0.91	0.91	0.81 $\pm$ 0.010	0.86 $\pm$ 0.009	0.81 $\pm$ 0.008
3rd Round	0.87	0.88	0.84	0.70 $\pm$ 0.015	0.89 $\pm$ 0.004	0.74 $\pm$ 0.014	0.96	0.91	0.92	0.84 $\pm$ 0.008	0.87 $\pm$ 0.015	0.83 $\pm$ 0.009
4th Round	0.85	0.88	0.83	0.73 $\pm$ 0.022	0.90 $\pm$ 0.013	0.77 $\pm$ 0.020	0.96	0.92	0.92	0.86 $\pm$ 0.006	0.87 $\pm$ 0.005	0.85 $\pm$ 0.004

Table 2: Pairwise statistical comparison of successive refinement rounds of IntentTrace for precision, recall and F1, based on the results in Table 1. Each round is compared against the one immediately before it. (S), (M) and (L) denote small, medium and large effect size. Yellow cells highlight comparisons where the newer round shows a statistically significant improvement over the previous round.

Dataset	LLM	Metric	1st Round vs No Refinement		2nd Round vs 1st Round		3rd Round vs 2nd Round		4th Round vs 3rd Round	
			p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$
Industry	Gemini	Precision	2.67E-33	0.78 (L)	2.80E-14	0.66 (M)	3.80E-05	0.57 (S)	0.71	0.51
		Recall	1.46E-25	0.71 (L)	1.25E-05	0.57 (S)	0.06	0.54	0.06	0.54
		F1	1.59E-32	0.74 (L)	1.60E-13	0.64 (M)	0.01	0.54	0.31	0.53
	Qwen	Precision	6.13E-120	0.71 (M)	1.72E-50	0.59 (S)	2.36E-22	0.55	1.81E-10	0.53
		Recall	1.12E-100	0.69 (M)	1.51E-13	0.58 (S)	6.08E-06	0.53	0.28	0.51
		F1	3.26E-123	0.70 (M)	2.88E-46	0.60 (S)	2.55E-23	0.55	4.76E-10	0.53
PAGED	Gemini	Precision	1.48E-44	0.76 (L)	5.12E-19	0.62 (S)	6.72E-06	0.54	0.34	0.51
		Recall	4.82E-35	0.69 (M)	2.12E-13	0.58 (S)	0.77	0.51	0.12	0.51
		F1	3.06E-43	0.73 (L)	1.36E-21	0.61 (S)	4.68E-03	0.53	0.04	0.51
	Qwen	Precision	1.06E-165	0.71 (L)	1.10E-48	0.60 (S)	1.13E-11	0.53	5.30E-06	0.52
		Recall	1.17E-124	0.67 (M)	1.79E-12	0.55	0.02	0.51	0.37	0.51
		F1	5.80E-163	0.70 (M)	1.76E-46	0.59 (S)	1.14E-10	0.53	2.06E-05	0.52

Table 3: Mean precision (P), recall (R), and F_1 after multiple rounds of refinement using IntentTrace under the ground-truth-based measures of LADEX and AbsCon. Structurally or logically unsound models score zero, as in Table 1. Bold marks the highest P, R, and F_1 within each measure-dataset-LLM setting. The values averaged over more than one run are provided with their standard deviation values.

Scorer	Refinement	Industry						Paged					
		Gemini			Qwen			Gemini			Qwen		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1
AbsCon	No Refinement	0.27 $\pm$ 0.000	0.40 $\pm$ 0.000	0.32 $\pm$ 0.000	0.27 $\pm$ 0.000	0.40 $\pm$ 0.000	0.32 $\pm$ 0.000	0.35 $\pm$ 0.000	0.36 $\pm$ 0.000	0.35 $\pm$ 0.000	0.35 $\pm$ 0.000	0.36 $\pm$ 0.000	0.35 $\pm$ 0.000
	1st Round	0.56	0.79	0.65	0.51 $\pm$ 0.008	0.71 $\pm$ 0.012	0.59 $\pm$ 0.010	0.68	0.65	0.65	0.63 $\pm$ 0.007	0.58 $\pm$ 0.007	0.59 $\pm$ 0.007
	2nd Round	0.60	0.84	0.69	0.54 $\pm$ 0.017	0.74 $\pm$ 0.021	0.62 $\pm$ 0.018	0.71	0.67	0.68	0.67 $\pm$ 0.006	0.60 $\pm$ 0.004	0.62 $\pm$ 0.005
	3rd Round	0.60	0.85	0.69	0.57 $\pm$ 0.005	0.76 $\pm$ 0.008	0.64 $\pm$ 0.006	0.71	0.66	0.67	0.68 $\pm$ 0.007	0.60 $\pm$ 0.007	0.63 $\pm$ 0.007
	4th Round	0.59	0.84	0.69	0.57 $\pm$ 0.005	0.76 $\pm$ 0.007	0.64 $\pm$ 0.006	0.71	0.66	0.67	0.69 $\pm$ 0.005	0.60 $\pm$ 0.006	0.63 $\pm$ 0.005
LADEX (B-MATCH)	No Refinement	0.43 $\pm$ 0.000	0.41 $\pm$ 0.000	0.41 $\pm$ 0.000	0.43 $\pm$ 0.000	0.41 $\pm$ 0.000	0.41 $\pm$ 0.000	0.42 $\pm$ 0.000	0.45 $\pm$ 0.000	0.42 $\pm$ 0.000	0.42 $\pm$ 0.000	0.45 $\pm$ 0.000	0.42 $\pm$ 0.000
	1st Round	0.82	0.70	0.72	0.78 $\pm$ 0.014	0.68 $\pm$ 0.009	0.70 $\pm$ 0.010	0.88	0.88	0.85	0.80 $\pm$ 0.014	0.80 $\pm$ 0.007	0.77 $\pm$ 0.010
	2nd Round	0.90	0.70	0.76	0.84 $\pm$ 0.031	0.66 $\pm$ 0.020	0.70 $\pm$ 0.025	0.93	0.89	0.89	0.85 $\pm$ 0.006	0.81 $\pm$ 0.010	0.81 $\pm$ 0.007
	3rd Round	0.90	0.69	0.74	0.88 $\pm$ 0.006	0.65 $\pm$ 0.009	0.71 $\pm$ 0.006	0.93	0.90	0.89	0.87 $\pm$ 0.010	0.81 $\pm$ 0.009	0.82 $\pm$ 0.008
	4th Round	0.90	0.69	0.75	0.89 $\pm$ 0.008	0.64 $\pm$ 0.008	0.71 $\pm$ 0.007	0.93	0.89	0.88	0.88 $\pm$ 0.012	0.81 $\pm$ 0.012	0.82 $\pm$ 0.013

Table 4: Pairwise statistical comparison of successive refinement rounds of IntentTrace for precision, recall and F1 under the ground-truth-based measure of AbsCon, based on the results in Table 3. Each round is compared against the one immediately before it. (S), (M) and (L) denote small, medium and large effect size. Yellow cells highlight comparisons where the newer round shows a statistically significant improvement over the previous round.

Dataset	LLM	Metric	1st Round vs No Refinement		2nd Round vs 1st Round		3rd Round vs 2nd Round		4th Round vs 3rd Round	
			p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$
Industry	Gemini	Precision	1.36E-31	0.77 (L)	9.79E-03	0.54	0.02	0.51	0.93	0.50
		Recall	1.75E-23	0.73 (L)	0.76	0.53	0.76	0.50	0.84	0.49
		F1	3.92E-29	0.77 (L)	9.79E-03	0.53	0.07	0.50	0.84	0.49
	Qwen	Precision	3.65E-131	0.73 (L)	3.89E-32	0.54	5.33E-17	0.53	1.97E-09	0.51
		Recall	7.08E-64	0.68 (M)	0.01	0.51	0.06	0.50	5.50E-04	0.50
		F1	2.21E-104	0.72 (L)	5.36E-08	0.53	3.47E-06	0.52	0.03	0.50
PAGED	Gemini	Precision	6.32E-40	0.73 (L)	8.88E-03	0.52	0.14	0.50	0.74	0.50
		Recall	2.86E-36	0.69 (M)	5.22E-03	0.51	0.71	0.49	0.01	0.49
		F1	5.96E-38	0.70 (M)	5.22E-03	0.52	0.13	0.49	0.03	0.49
	Qwen	Precision	2.78E-155	0.69 (M)	3.56E-27	0.54	1.31E-06	0.51	2.18E-03	0.51
		Recall	5.56E-92	0.64 (M)	0.10	0.51	8.40E-03	0.50	0.03	0.50
		F1	4.93E-113	0.66 (M)	9.90E-06	0.52	0.65	0.50	0.95	0.50

Table 5: Pairwise statistical comparison of successive refinement rounds of IntentTrace for precision, recall and F1 under the ground-truth-based measure of LADEX, based on the results in Table 3. Each round is compared against the one immediately before it. (S), (M) and (L) denote small, medium and large effect size. Yellow cells highlight comparisons where the newer round shows a statistically significant improvement over the previous round.

Dataset	LLM	Metric	1st Round vs No Refinement		2nd Round vs 1st Round		3rd Round vs 2nd Round		4th Round vs 3rd Round	
			p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$
Industry	Gemini	Precision	6.82E-25	0.72 (L)	1.02E-06	0.57 (S)	0.96	0.49	0.96	0.51
		Recall	2.49E-16	0.68 (M)	0.46	0.49	0.08	0.49	0.34	0.51
		F1	9.45E-18	0.68 (M)	0.67	0.51	0.03	0.48	0.16	0.51
	Qwen	Precision	6.49E-101	0.70 (M)	1.29E-08	0.55	6.30E-06	0.53	0.16	0.51
		Recall	4.21E-69	0.66 (M)	7.29E-03	0.47	0.73	0.49	0.05	0.48
		F1	3.11E-78	0.66 (M)	0.31	0.49	0.71	0.50	0.22	0.49
PAGED	Gemini	Precision	5.88E-40	0.75 (L)	2.18E-07	0.54	0.16	0.51	0.17	0.49
		Recall	1.39E-36	0.73 (L)	0.26	0.49	0.34	0.51	0.16	0.49
		F1	3.17E-39	0.75 (L)	1.94E-03	0.52	0.16	0.51	0.13	0.49
	Qwen	Precision	1.43E-144	0.70 (M)	9.66E-10	0.54	2.27E-03	0.52	0.50	0.51
		Recall	1.19E-122	0.68 (M)	0.05	0.50	0.65	0.49	0.78	0.49
		F1	7.97E-140	0.70 (M)	1.39E-05	0.53	0.65	0.50	0.89	0.50

Table 6: Paired statistical comparison of precision, recall, and F_1 between IntentTrace with Qwen (averaged over five runs) and IntentTrace with Gemini, evaluated after a single round and after Qwen’s fourth round. $\hat{A}_{12} > 0.5$ favors Qwen; (S), (M), and (L) denote small, medium, and large effect sizes. Yellow and blue cells highlight a significant advantage for Qwen and Gemini, respectively.

Dataset	Metric	Qwen 1st Round vs Gemini 1st Round		Qwen 4th Round vs Gemini 1st Round	
		p	$\hat{A}_{12}$	p	$\hat{A}_{12}$
Industry	Precision	3.48E-17	0.33 (M)	0.30	0.50
	Recall	0.13	0.42 (S)	3.65E-04	0.56 (S)
	F1	5.48E-11	0.37 (S)	4.91E-03	0.54
PAGED	Precision	1.05E-21	0.32 (M)	0.52	0.47
	Recall	1.16E-03	0.42 (S)	0.41	0.50
	F1	1.30E-15	0.37 (S)	0.52	0.50

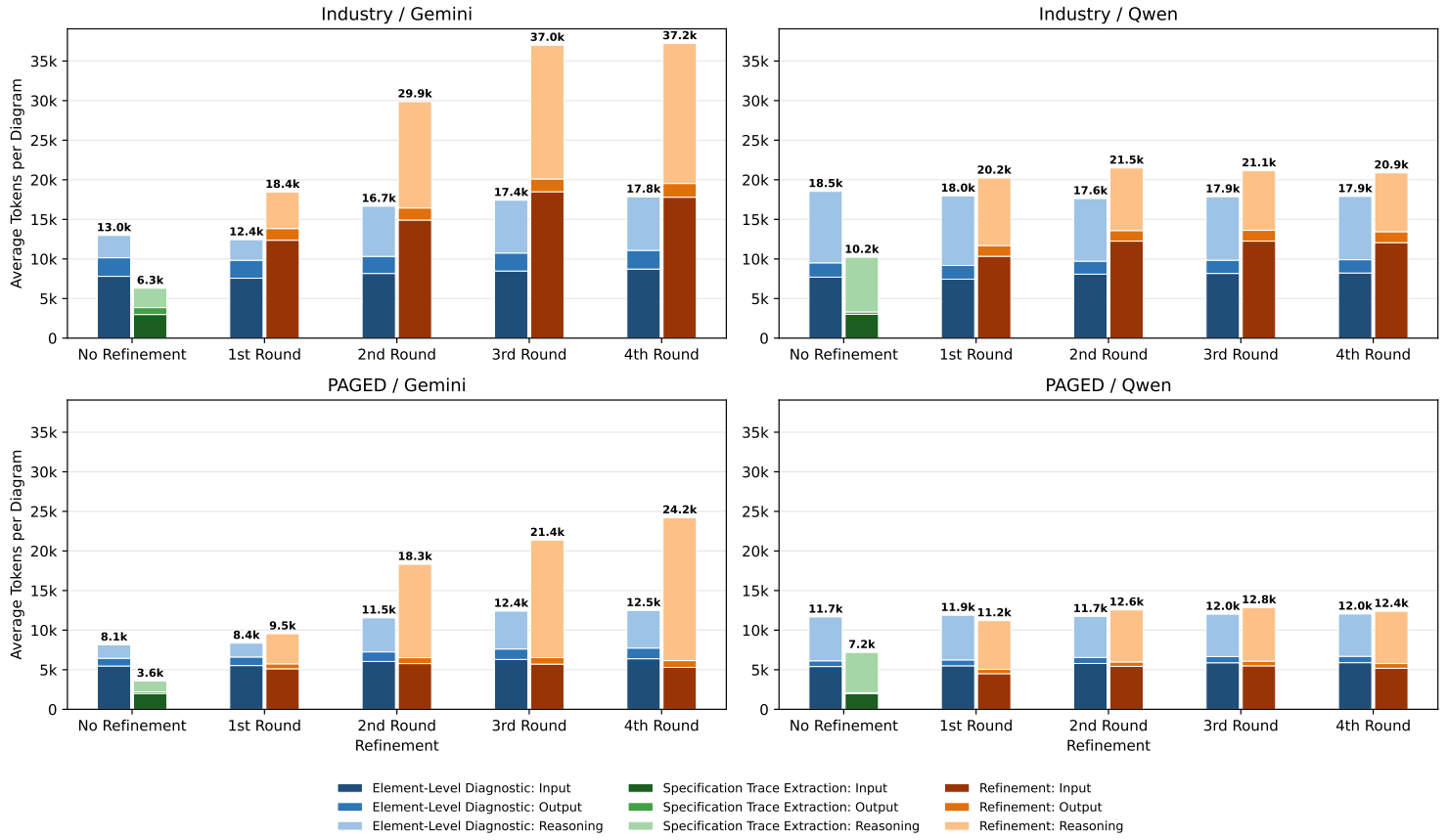


Figure 1: Average tokens for one model passing through each refinement round, stacked by token kind (input, output, reasoning) and grouped by pipeline stage. Each stage is averaged over only the models that issued that stage’s call. One panel per dataset–LLM setting on a shared y-axis. Bar labels give the stage total in thousands.